

# Sentiment Analysis of Sephora Products Reviews

## 1. Introduction

- Purpose of the Project

The primary purpose of this project is to perform **sentiment analysis** on customer reviews from the Sephora products dataset. The goal is to **classify whether a review reflects a positive or negative sentiment**, using the `'is_recommended'` column as the target variable. Additionally, the project aims to explore trends and patterns in reviews, offering deeper insights into customer preferences and feedback.

- Main Assumptions

- Negative reviews are longer on average than positive reviews.
- More expensive brands are associated with a higher proportion of positive reviews.

- Research Areas

- Investigating trends in positive and negative reviews over time.
- Identifying which product categories or brands receive more positive or negative reviews.
- Analyzing the relationship between review length and sentiment.
- Exploring the impact of brand pricing on customer satisfaction.

## 2. Data

- Description of the Dataset

The dataset is sourced from [Kaggle](#) and contains reviews of [Sephora products](#). Key attributes include:

- `'review_text'`: Textual content of customer reviews.
- `'is_recommended'`: Binary indicator of whether the customer recommends the product (1 = true, 0 = false).
- `'submission_time'`: Date the review was posted, useful for analyzing trends over time.
- `'rating'`: Numerical rating (1–5) provided by the customer.
- `'total_feedback_count'`: Number of feedback responses received for the review.
- Additional product-level metadata, such as product ID, brand, and category.

## 3. Modeling

- Algorithms to be Used

- Logistic Regression
  - Naive Bayes Classifier
  - Random Forest Classifier
  - Gradient Boosting
- Feature Representation:
- Text data will be vectorized using **TF-IDF**.

#### **4. Summary**

- Expected Outcomes
- Accurately classify customer reviews as positive or negative based on sentiment.
  - Identify trends in sentiment over time.
  - Highlight key differences in sentiment across product categories and brands.
  - Validate or refute the assumptions about review length and brand pricing's influence on sentiment.